

# Cointegrated Currency Linkages During Crises

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## Model Setup

We now present a regime-dependent factor model linking the long-run dynamics of  $N$  non-USD exchange rates to a set of nonstationary global factors. The U.S. is the numeraire. Let the spot exchange rate be

$$S_{i,t} = \text{units of currency } i \text{ per USD} \quad (i = 1, \dots, N),$$

so an increase in  $S_{i,t}$  indicates USD appreciation against currency  $i$ . Define

$$X_{i,t} = \log S_{i,t}, \quad X_t = (X_{1,t}, \dots, X_{N,t})'.$$

## Explanation of Model Setup

This part is setting up the notation and orientation of the exchange rate system before introducing the stochastic discount factor (SDF) model. We unpack it carefully below.

### Choice of Numeraire

The U.S. dollar (USD) is chosen as the numeraire. This means that all exchange rates are expressed in terms of how many units of a foreign currency equal one USD.

### Definition of $S_{i,t}$

$$S_{i,t} = \text{units of currency } i \text{ per USD}, \quad (i = 1, \dots, N).$$

- If  $i = 1$  corresponds to the euro, then  $S_{1,t}$  is EUR/USD (e.g., 0.92 EUR per USD).
- If  $i = 2$  corresponds to the yen, then  $S_{2,t}$  is JPY/USD (e.g., 150 JPY per USD).

By construction:

- An increase in  $S_{i,t}$  means it takes more units of currency  $i$  to buy one USD, implying that the USD has appreciated relative to currency  $i$ .
- A decrease in  $S_{i,t}$  means it takes fewer units of currency  $i$  per USD, implying that the USD has depreciated relative to currency  $i$ .

## Logarithmic Transformation

The log exchange rate is defined as

$$X_{i,t} = \log S_{i,t}.$$

- This transformation is standard, as log-prices simplify many econometric properties such as unit-root testing, cointegration analysis, and VECM estimation.
- Log-differences,  $\Delta X_{i,t} = X_{i,t} - X_{i,t-1}$ , correspond approximately to percentage changes (returns).

## Vector Notation

Finally, all  $N$  log exchange rates are stacked into a single vector:

$$X_t = (X_{1,t}, \dots, X_{N,t})',$$

allowing the model to be expressed compactly in matrix and vector form.

## Why Log-Differences Approximate Returns

Consider the log exchange rate

$$X_{i,t} = \log S_{i,t},$$

where  $S_{i,t}$  is the spot exchange rate (units of currency  $i$  per USD). The log-difference is then

$$\Delta X_{i,t} = X_{i,t} - X_{i,t-1} = \log S_{i,t} - \log S_{i,t-1}.$$

By properties of the logarithm,

$$\Delta X_{i,t} = \log \left( \frac{S_{i,t}}{S_{i,t-1}} \right).$$

Now, the gross return from  $t-1$  to  $t$  is

$$R_{i,t} = \frac{S_{i,t} - S_{i,t-1}}{S_{i,t-1}} = \frac{S_{i,t}}{S_{i,t-1}} - 1.$$

Hence, the log-difference can be expressed as

$$\Delta X_{i,t} = \log(1 + R_{i,t}).$$

For small returns (as is typically the case in daily or weekly foreign exchange data), the first-order Taylor approximation of the logarithm applies:

$$\log(1 + R_{i,t}) \approx R_{i,t}.$$

Therefore,

$$\Delta X_{i,t} \approx R_{i,t},$$

showing that log-differences of exchange rates correspond approximately to percentage changes (returns). This property makes log-prices particularly convenient for econometric analysis, as they linearize compounding and preserve additive time-series properties.

## 2. Pricing Kernel (SDF)

We price assets in USD using a log-normal stochastic discount factor (SDF):

$$\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}, \quad (1)$$

where  $r_t$  is the USD short rate,  $\xi_{t+1}$  is the SDF innovation, and  $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$  by construction.

**Multi-factor global structure.** In regime  $r$ , the innovation admits  $q_r$  nonstationary global factors:

$$\xi_{t+1} = \lambda_r' g_{t+1} + v_{t+1}, \quad \mathbb{E}_t[g_{t+1}v_{t+1}] = 0, \quad (2)$$

where:

- $g_t = (g_{1,t}, \dots, g_{q_r,t})'$  are  $q_r$  independent  $I(1)$  global factors, e.g., USD funding stress, global risk appetite, commodity-price shocks.
- $\lambda_r \in \mathbb{R}^{q_r}$  are SDF-level factor loadings in regime  $r$ .
- $v_{t+1} \sim \mathcal{N}(0, \sigma_v^2)$  is an idiosyncratic USD pricing-kernel shock orthogonal to  $g_{t+1}$ .

## Explanation of Pricing Kernel (SDF)

This section introduces the stochastic discount factor (SDF), which is the core object in modern asset pricing. The SDF—also called the pricing kernel—is the random variable that converts future payoffs into present values. Formally, the price of any asset with payoff  $D_{t+1}$  is

$$P_t = \mathbb{E}_t[M_{t+1}D_{t+1}],$$

so the dynamics of  $M_{t+1}$  directly determine equilibrium asset prices, including exchange rates.

### Log-Normal Specification

The paper assumes that the USD SDF follows a log-normal process of the form:

$$\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}, \quad (1)$$

where:

- $r_t$  is the USD short rate (the one-period risk-free interest rate).
- $\varsigma_r^2$  is the conditional variance of SDF innovations in regime  $r$ .
- $\xi_{t+1}$  is the SDF innovation (random shock).

The term  $-\frac{1}{2}\varsigma_r^2$  is a Jensen correction ensuring that  $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$ , so the risk-free asset earns exactly the short rate by construction. This condition enforces internal consistency with the no-arbitrage requirement.

## The Jensen Correction

We recall that the mean of a standard normal random variable  $\xi \sim N(0, 1)$  is equal to zero. However, in the present case we are not evaluating  $E[\xi]$ , but rather the expectation of an exponential transformation of  $\xi$ :

$$E[e^{a\xi}].$$

By definition,

$$E[e^{a\xi}] = \int_{-\infty}^{\infty} e^{ax} f_{\xi}(x) dx,$$

where  $f_{\xi}(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$  is the density of the standard normal distribution. Substituting this density yields

$$E[e^{a\xi}] = \int_{-\infty}^{\infty} e^{ax} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(ax - \frac{1}{2}x^2) dx.$$

We now complete the square inside the exponent. Observe that

$$ax - \frac{1}{2}x^2 = -\frac{1}{2}(x^2 - 2ax) = -\frac{1}{2}((x - a)^2 - a^2) = -\frac{1}{2}(x - a)^2 + \frac{1}{2}a^2.$$

Substituting back, we obtain

$$E[e^{a\xi}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(-\frac{1}{2}(x - a)^2 + \frac{1}{2}a^2) dx = e^{\frac{1}{2}a^2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(-\frac{1}{2}(x - a)^2) dx.$$

The remaining integral is equal to one, since it represents the total probability under a  $N(a, 1)$  distribution. Hence,

$$E[e^{a\xi}] = e^{\frac{1}{2}a^2}.$$

This shows that the expectation is not zero, because we are averaging the nonlinear function  $e^{a\xi}$ , which is strictly positive and has upward skew relative to the mean of  $\xi$ .

The stochastic discount factor is specified as

$$M_{t+1} = \exp(-rt - \frac{1}{2}\varsigma_r^2 + \varsigma_r \xi_{t+1}), \quad \xi_{t+1} \sim N(0, 1).$$

Taking expectations,

$$\mathbb{E}_t[M_{t+1}] = e^{-rt - \frac{1}{2}\varsigma_r^2} \mathbb{E}_t[e^{\varsigma_r \xi_{t+1}}].$$

Since  $\mathbb{E}[e^{a\xi}] = e^{\frac{1}{2}a^2}$  for  $\xi \sim N(0, 1)$ ,

$$\mathbb{E}_t[M_{t+1}] = e^{-rt - \frac{1}{2}\varsigma_r^2} e^{\frac{1}{2}\varsigma_r^2} = e^{-rt}.$$

Hence the  $-\frac{1}{2}\varsigma_r^2$  term is exactly the Jensen correction that ensures the no-arbitrage condition holds.

## Fundamental Pricing Relation

The present value of \$1 to be received at time  $t$ , under continuous discounting at rate  $r$ , is:

$$p(0, t) = e^{-rt}.$$

More generally, for a payoff of  $X$  dollars at time  $t$ , the present value is:

$$p(0, t; X) = X e^{-rt}.$$

## Structure of Innovations

The innovation  $\xi_{t+1}$  is then decomposed into global and idiosyncratic components. In regime  $r$ :

$$\xi_{t+1} = \lambda_r' g_{t+1} + \nu_{t+1}, \quad \mathbb{E}_t[g_{t+1}\nu_{t+1}] = 0, \quad (2)$$

where:

- $g_{t+1}$  is a  $q_r \times 1$  vector of nonstationary global factors (e.g., dollar funding stress, global risk appetite, or commodity price shocks). These factors are integrated processes that introduce persistent stochastic trends into asset prices.
- $\lambda_r$  is a vector of factor loadings that governs the sensitivity of the USD SDF to each global factor in regime  $r$ .
- $\nu_{t+1}$  is an idiosyncratic shock specific to the USD SDF, assumed to be orthogonal to the global factors  $g_{t+1}$ .

## Economic Interpretation

Equation (2) shows that the dynamics of the SDF are driven by two distinct sources:

- A **systematic component**,  $\lambda_r' g_{t+1}$ , which reflects exposure to common global forces that simultaneously influence many currencies.
- An **idiosyncratic component**,  $\nu_{t+1}$ , which captures USD-specific shocks uncorrelated with the global drivers.

This decomposition is crucial because the nonstationary global factors  $g_{t+1}$  are shared across currencies. As a result, multiple exchange rates inherit common stochastic trends, which in turn generate long-run equilibrium linkages that can be studied using cointegration techniques. At the same time, the presence of  $\nu_{t+1}$  allows for short-run deviations and idiosyncratic noise that do not affect long-run relationships.

## Interpretation of Regime $r$

The phrase *regime  $r$*  plays a central role in the paper’s setup. In time-series econometrics and asset pricing, a **regime** refers to a period during which the economy or market is governed by a specific set of dynamics or parameters. Different regimes capture structural changes in volatility, correlations, or overall market behavior. Intuitively, one can think of this as the model “switching” between different states of the world.

## Application in this paper

In the context of this paper, *regime  $r$*  indexes the macro-financial environment of the foreign exchange market. Each regime corresponds to its own parameterization of the stochastic discount factor (SDF). For instance:

- **Pre-crisis regime:** stable conditions with normal liquidity and lower volatility.
- **Crisis regime:** periods such as the Global Financial Crisis (2007–2009) or the COVID-19 pandemic (2020–2022), characterized by higher volatility, tighter funding conditions, and reduced autonomy across currencies.
- **Post-crisis regime:** recovery phases, where parameters gradually return toward more stable values.

Thus, when the paper writes terms such as

$$\zeta_r^2,$$

it emphasizes that the conditional variance of SDF innovations depends explicitly on the prevailing regime  $r$ .

### Why regime-dependence matters

Several model parameters, including volatility ( $\varsigma_r^2$ ) or autonomy ( $\psi_{i,r}$ ), are regime-specific. This structure allows the model to capture why exchange rate dynamics differ across environments such as the GFC, the COVID-19 crisis, or post-crisis normalization. In particular:

- During the GFC, autonomy compresses and cointegration across exchange rates strengthens.
- During COVID-19, autonomy expands and cointegration weakens.

Each factor follows a random walk with regime-dependent drift and volatility:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_{k,r}} \varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1), \quad k = 1, \dots, q_r. \quad (3)$$

The total SDF innovation variance is:

$$\varsigma_r^2 = \lambda_r' \Sigma_{g,r} \lambda_r + \sigma_v^2, \quad \Sigma_{g,r} = \text{diag}(\sigma_{g_{1,r}}^2, \dots, \sigma_{g_{q_r,r}}^2).$$

## Factor Dynamics and SDF Variance

Each factor  $g_{k,t}$  evolves according to a random walk with regime-dependent drift and volatility:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_{k,r}} \varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1), \quad k = 1, \dots, q_r.$$

This formulation implies that:

- $\mu_{k,r}$  is the drift of factor  $k$  in regime  $r$ , determining its expected incremental change per period.
- $\sigma_{g_{k,r}}$  is the volatility of factor  $k$  in regime  $r$ , scaling the standard normal shock  $\varepsilon_{k,t+1}$ .
- The distribution of innovations is regime-dependent, meaning that both average growth rates and volatility levels vary depending on whether the market is in a pre-crisis, crisis, or post-crisis environment.

The total stochastic discount factor (SDF) innovation variance in regime  $r$  is defined as:

$$\varsigma_r^2 = \lambda_r' \Sigma_{g,r} \lambda_r + \sigma_v^2, \quad \Sigma_{g,r} = \text{diag}(\sigma_{g_{1,r}}^2, \dots, \sigma_{g_{q_r,r}}^2).$$

Breaking down each component:

- $\varsigma_r^2$ : the overall variance of SDF innovations in regime  $r$ .
- $\lambda_r$ : a  $q_r \times 1$  column vector of regime-specific factor loadings. Each element  $\lambda_{k,r}$  measures how strongly the SDF reacts to shocks in the corresponding factor  $g_{k,r}$ .
- $\lambda_r'$ : the transpose of  $\lambda_r$ , which is a  $1 \times q_r$  row vector. The use of both  $\lambda_r$  and  $\lambda_r'$  is necessary to construct a quadratic form that collapses to a scalar variance. Specifically:

$$(1 \times q_r) \cdot (q_r \times q_r) \cdot (q_r \times 1) = (1 \times 1),$$

ensuring that the expression  $\lambda_r' \Sigma_{g,r} \lambda_r$  evaluates to a scalar.

- $\Sigma_{g,r}$ : a diagonal matrix containing factor innovation variances,

$$\Sigma_{g,r} = \begin{bmatrix} \sigma_{g_1,r}^2 & 0 & \cdots & 0 \\ 0 & \sigma_{g_2,r}^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{g_{q_r},r}^2 \end{bmatrix}.$$

This structure reflects the assumption that factor shocks are independent across  $k$ .

- $\lambda_r' \Sigma_{g,r} \lambda_r$ : the quadratic form capturing the contribution of factor-driven risk. By matrix multiplication, this expands to

$$\lambda_r' \Sigma_{g,r} \lambda_r = \sum_{k=1}^{q_r} \lambda_{k,r}^2 \sigma_{g_k,r}^2,$$

which shows explicitly that each factor's variance  $\sigma_{g_k,r}^2$  is weighted by the square of its factor loading  $\lambda_{k,r}$ , and then summed across all factors. This aggregation gives the systematic component of SDF variance in regime  $r$ .

- $\sigma_\nu^2$ : an additional variance component representing idiosyncratic shocks to the SDF that are not captured by the observed factors. It ensures that the SDF retains residual unpredictability even after accounting for factor dynamics.

Thus,  $\varsigma_r^2$  combines both systematic factor risk and idiosyncratic noise:

$$\varsigma_r^2 = \underbrace{\lambda_r' \Sigma_{g,r} \lambda_r}_{\text{systematic factor-driven risk}} + \underbrace{\sigma_\nu^2}_{\text{idiosyncratic risk}}.$$

### 3. Payoffs, No-Arbitrage and Affine Solution

A one-period claim pays  $D_{i,t+1}$  units of currency  $i$ ; its USD price is

$$P_{i,t} = \mathbb{E}_t \left[ M_{t+1} \frac{D_{i,t+1}}{S_{i,t+1}} \right]. \quad (4)$$

For a pure FX claim delivering one unit of currency  $i$  (i.e.  $D_{i,t+1} = 1$ ), replication implies

## Pricing of Foreign Currency Payoffs

Consider a one-period claim that pays  $D_{i,t+1}$  units of foreign currency  $i$  at time  $t+1$ . We want to compute its USD price today, denoted  $P_{i,t}$ .

### Converting Payoffs into USD

The claim pays  $D_{i,t+1}$  units of foreign currency  $i$ . To value this payoff in USD, we use the spot exchange rate  $S_{i,t+1}$ , defined as

$$S_{i,t+1} = \frac{\text{currency } i}{\text{USD}}.$$

The reciprocal therefore has units

$$\frac{1}{S_{i,t+1}} = \frac{\text{USD}}{\text{currency } i}.$$

Hence, the payoff in USD is

$$\text{Payoff in USD at time } t+1 = D_{i,t+1} \times \frac{1}{S_{i,t+1}}.$$

Checking units explicitly:

$$\underbrace{D_{i,t+1}}_{[\text{currency } i]} \times \underbrace{\frac{1}{S_{i,t+1}}}_{[\text{USD}]/[\text{currency } i]} = [\text{USD}].$$

Thus, dividing by  $S_{i,t+1}$  correctly converts the foreign-currency payoff into USD terms.

## Pricing via the Stochastic Discount Factor

By the no-arbitrage pricing equation, the USD price today equals the expected discounted value of the USD payoff:

$$P_{i,t} = \mathbb{E}_t \left[ M_{t+1} \cdot \frac{D_{i,t+1}}{S_{i,t+1}} \right],$$

where  $M_{t+1}$  is the stochastic discount factor (SDF).

## Special Case – Pure FX Claim

For a pure foreign exchange claim that delivers one unit of currency  $i$ , we set  $D_{i,t+1} = 1$ . In this case, the payoff is simply

$$\frac{1}{S_{i,t+1}},$$

and the USD price becomes

$$P_{i,t} = \mathbb{E}_t \left[ M_{t+1} \cdot \frac{1}{S_{i,t+1}} \right].$$

$P_{i,t} = 1/S_{i,t}$ . No-arbitrage gives:

$$\frac{1}{S_{i,t}} = \mathbb{E}_t \left[ M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

or equivalently,

$$1 = \mathbb{E}_t \left[ M_{t+1} \frac{S_{i,t}}{S_{i,t+1}} \right] = \mathbb{E}_t \left[ M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right], \quad (5)$$

our key FX pricing restriction.

The intuition behind the no-(riskless)-arbitrage condition is that the dollar price paid today for a future claim of one unit of foreign currency

$$\mathbb{E}_t \left[ M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

must equal the current dollar cost of acquiring that same unit of foreign currency,

$$\frac{1}{S_{i,t}},$$



If this equality does not hold, an investor could exploit the difference to earn a profit without risk. Any profit that involves bearing risk is not considered a riskless arbitrage profit and is therefore outside the scope of our model.

## Equation 5 Derivation

### Step 1: Setup

Consider a contract that delivers one unit of foreign currency  $i$  at time  $t + 1$ . Its value at time  $t$  in USD is denoted

$$P_{i,t} = \frac{1}{S_{i,t}}.$$

Here, the spot exchange rate is defined as

$$S_{i,t} = \frac{\text{currency } i}{\text{USD}}.$$

Thus,

$$\frac{1}{S_{i,t}} \sim \frac{\text{USD}}{\text{currency } i}.$$

Multiplying this by one unit of currency  $i$  gives a payoff denominated in USD.

### Step 2: No-Arbitrage Condition

No-arbitrage requires that the USD price today equals the expected, SDF-discounted payoff tomorrow:

$$\frac{1}{S_{i,t}} = \mathbb{E}_t \left[ M_{t+1} \cdot \frac{1}{S_{i,t+1}} \right].$$

### Step 3: Rearrangement

Multiplying numerator and denominator by  $S_{i,t}$  yields

$$1 = \mathbb{E}_t \left[ M_{t+1} \cdot \frac{S_{i,t}}{S_{i,t+1}} \right].$$

The term  $\frac{S_{i,t}}{S_{i,t+1}}$  represents the *gross return* on a foreign currency investment: buying one unit of foreign currency today and converting it back into USD tomorrow.

### Step 4: Unit Check

Explicitly,

$$\underbrace{\frac{1}{S_{i,t}}}_{\left[ \frac{\text{USD}}{\text{currency } i} \right]} = \underbrace{\mathbb{E}_t \left[ M_{t+1} \cdot \frac{1}{S_{i,t+1}} \right]}_{\left[ \text{unitless} \right] \times \left[ \frac{\text{USD}}{\text{currency } i} \right]}.$$

Both sides are expressed in USD, confirming dimensional consistency.

## Step 5: Economic Intuition

- **LHS:** The current USD cost of acquiring one unit of foreign currency is  $\frac{1}{S_{i,t}}$ .
- **RHS:** The expected USD value of receiving one unit of foreign currency at  $t+1$ , discounted by the SDF  $M_{t+1}$ .

Therefore, under no-arbitrage, the current USD cost of foreign currency equals the discounted expected future value of holding it. Any deviation would imply the existence of a riskless arbitrage opportunity.

### 4. Solving for Exchange-Rate Levels

Assuming (i) the SDF is log-normal as in (??)–(??), and (ii) fundamentals  $z_t$  ( $K \times 1$ ) are stationary and independent of the global-factor innovations, the log-linear solution takes the affine form

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}, \quad (6)$$

with  $u_{i,t}$  stationary, and

$$\mu_{i,r} = -\bar{r}_r - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \kappa_{i,r}, \quad \text{with } \kappa_{i,r} \equiv \frac{1}{2} [(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2] + \Delta(\Gamma'_i z_t + u_{i,t})$$

$$\text{Equation getting clipped here. I recommend breaking it up into multiple lines.} \quad (7)$$

$$\Gamma'_i z_t = \sum_{k=1}^K \gamma_{i,k} z_{k,t}, \quad z_t = (z_{1,t}, \dots, z_{K,t})' \text{ is stationary,} \quad (8)$$

$$u_{i,t} = X_{i,t} - \mu_{i,r} - \lambda'_{i,r} g_t - \Gamma'_i z_t, \quad u_{i,t} \text{ stationary.} \quad (9)$$

## Derivation of Equation 7

### Assumptions

$$g_{t+1} = g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, I), \quad \Sigma_{g,r} = \sigma_{g,r} \sigma_{g,r}^\top = \text{diag}(\sigma_{g1,r}^2, \dots, \sigma_{gqr,r}^2).$$

$$\log M_{t+1} = -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1}, \quad \varsigma_r^2 = \lambda_r^\top \Sigma_{g,r} \lambda_r + \sigma_v^2, \quad v_{t+1} \sim \mathcal{N}(0, \sigma_v^2).$$

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top z_t + u_{i,t}, \quad S_{i,t} := \Gamma_i^\top z_t + u_{i,t}, \quad \Delta S_{i,t} \text{ stationary with } \mathbb{E}_t[\Delta S_{i,t}] = 0.$$

### No Arbitrage

$$1 = \mathbb{E}_t[M_{t+1} e^{X_{i,t} - X_{i,t+1}}], \quad Y_{t+1} := \log M_{t+1} + X_{i,t} - X_{i,t+1}.$$

### Computing $X_{i,t} - X_{i,t+1}$

Write the affine form at  $t$  and  $t+1$

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + S_{i,t}, \quad X_{i,t+1} = \mu_{i,r} + \lambda_{i,r}^\top g_{t+1} + S_{i,t+1},$$

where  $S_{i,t} := \Gamma_i^\top z_t + u_{i,t}$ .

**Subtract the two expressions**

$$\begin{aligned}
X_{i,t} - X_{i,t+1} &= (\mu_{i,r} + \lambda_{i,r}^\top g_t + S_{i,t}) - (\mu_{i,r} + \lambda_{i,r}^\top g_{t+1} + S_{i,t+1}) \\
&= \lambda_{i,r}^\top (g_t - g_{t+1}) - (S_{i,t+1} - S_{i,t}) \quad (\text{the two } \mu_{i,r} \text{ cancel}) \\
&= -\lambda_{i,r}^\top (g_{t+1} - g_t) - \Delta S_{i,t},
\end{aligned}$$

where  $\Delta S_{i,t} := S_{i,t+1} - S_{i,t}$ .

**Expand  $\Delta S_{i,t}$**

$$\Delta S_{i,t} = \Gamma_i^\top (z_{t+1} - z_t) + (u_{i,t+1} - u_{i,t}) = \Gamma_i^\top \Delta z_t + \Delta u_{i,t}.$$

**Use the paper's factor law for  $g_{t+1}$**

$$\begin{aligned}
g_{t+1} - g_t &= \mu_r + \sigma_{g,r} \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, I). \\
X_{i,t} - X_{i,t+1} &= -\lambda_{i,r}^\top (\mu_r + \sigma_{g,r} \varepsilon_{t+1}) - \Delta S_{i,t} \\
&= -\lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta S_{i,t}.
\end{aligned}$$

**Assemble  $Y_{t+1}$**

**Define the object whose expectation we evaluate**

By the FX restriction  $1 = \mathbb{E}_t[M_{t+1} e^{X_{i,t} - X_{i,t+1}}]$ , set

$$Y_{t+1} := \log M_{t+1} + X_{i,t} - X_{i,t+1}.$$

**Substitute the SDF and the difference from the previous subsection**

Using  $\log M_{t+1} = -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1}$  and  $X_{i,t} - X_{i,t+1} = -\lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta S_{i,t}$ , we get

$$\begin{aligned}
Y_{t+1} &= \left( -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1} \right) + \left( -\lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta S_{i,t} \right) \\
&= -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_{t+1} - \lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}.
\end{aligned}$$

**Expand  $g_{t+1}$  using the factor law**

Since  $g_{t+1} = g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}$ ,

$$-\lambda_r^\top g_{t+1} = -\lambda_r^\top (g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}) = -\lambda_r^\top g_t - \lambda_r^\top \mu_r - \lambda_r^\top \sigma_{g,r} \varepsilon_{t+1}.$$

**Substitute and collect like terms**

$$\begin{aligned}
Y_{t+1} &= -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_t - \lambda_r^\top \mu_r - \lambda_r^\top \sigma_{g,r} \varepsilon_{t+1} \\
&\quad - \lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t} \\
&= -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r - (\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}.
\end{aligned}$$

**Identify deterministic vs. shock terms**

$$\underbrace{-r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r}_{\text{deterministic/state part}} + \underbrace{-(\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}}_{\text{innovations}}.$$

**Mean/variance and log-normal expectation**

$$\begin{aligned}
\mathbb{E}_t[Y_{t+1}] &= -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r, \\
\text{Var}_t(Y_{t+1}) &= (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta S_{i,t}).
\end{aligned}$$

## Compute $\text{Var}_t(Y_{t+1})$

Start from the assembled  $Y_{t+1}$

From the previous subsection we have

$$Y_{t+1} = \underbrace{-r_t - \frac{1}{2}\zeta_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r}_{\text{deterministic, } \mathcal{F}_t\text{-measurable}} - \underbrace{(\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}}_{(a)} - \underbrace{v_{t+1}}_{(b)} - \underbrace{\Delta S_{i,t}}_{(c)}.$$

## Identify the random innovations and their properties

By assumption in regime  $r$ :

$$\varepsilon_{t+1} \sim \mathcal{N}(0, I), \quad v_{t+1} \sim \mathcal{N}(0, \sigma_v^2), \quad \Delta S_{i,t} \text{ is stationary with finite variance.}$$

Moreover,  $\varepsilon_{t+1}$  is *independent* of  $v_{t+1}$  and of the stationary block, and all have zero conditional mean given  $\mathcal{F}_t$ . Also  $\Sigma_{g,r} = \sigma_{g,r} \sigma_{g,r}^\top = \text{diag}(\sigma_{g1,r}^2, \dots, \sigma_{gq_r,r}^2)$ .

## Drop the deterministic part for the variance

Let

$$A_t := -r_t - \frac{1}{2}\zeta_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r \quad (\mathcal{F}_t\text{-measurable}),$$

and write

$$Y_{t+1} = A_t - (\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}.$$

Since  $A_t$  is  $\mathcal{F}_t$ -measurable,  $\text{Var}_t(A_t) = 0$ . Hence

$$\text{Var}_t(Y_{t+1}) = \text{Var}_t\left(-(\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}\right).$$

## Use variance-of-sum with independence

Because  $\varepsilon_{t+1}$ ,  $v_{t+1}$ , and  $\Delta S_{i,t}$  are mutually independent (and each has conditional mean zero),

$$\text{Var}_t(Y_{t+1}) = \text{Var}_t((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}) + \text{Var}_t(v_{t+1}) + \text{Var}_t(\Delta S_{i,t}).$$

Recall, for any scalar  $c$  and random variable  $X$ ,

$$\text{Var}(cX) = c^2 \text{Var}(X),$$

So  $\text{Var}(x) = \text{Var}(-x)$ .

## Important Variance Property

The property that  $\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z)$  for independent random variables is fundamental in probability theory and statistics.

## General Variance Formula

For any random variables  $X$ ,  $Y$ , and  $Z$ , the general formula for the variance of their sum is:

$$\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z) + 2\text{Cov}(X, Y) + 2\text{Cov}(X, Z) + 2\text{Cov}(Y, Z)$$

## Why Independence Eliminates Covariance Terms

Independence means that knowing the value of one variable gives no information about another. Mathematically, this translates to:

- If  $X$  and  $Y$  are independent, then  $E[XY] = E[X]E[Y]$
- Covariance is defined as:  $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- Therefore, for independent  $X$  and  $Y$ :

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0$$

## Mathematical Proof

Starting with the definition of variance:

$$\text{Var}(X + Y + Z) = E[(X + Y + Z)^2] - [E(X + Y + Z)]^2$$

When we expand  $(X + Y + Z)^2$ , we get:

$$(X + Y + Z)^2 = X^2 + Y^2 + Z^2 + 2XY + 2XZ + 2YZ$$

Taking expectations:

$$E[(X + Y + Z)^2] = E[X^2] + E[Y^2] + E[Z^2] + 2E[XY] + 2E[XZ] + 2E[YZ]$$

For the second term:

$$\begin{aligned} [E(X + Y + Z)]^2 &= [E[X] + E[Y] + E[Z]]^2 \\ &= E[X]^2 + E[Y]^2 + E[Z]^2 + 2E[X]E[Y] + 2E[X]E[Z] + 2E[Y]E[Z] \end{aligned}$$

Subtracting equation (7) from equation (6):

$$\begin{aligned} \text{Var}(X + Y + Z) &= E[X^2] + E[Y^2] + E[Z^2] + 2E[XY] + 2E[XZ] + 2E[YZ] \\ &\quad - E[X]^2 - E[Y]^2 - E[Z]^2 - 2E[X]E[Y] - 2E[X]E[Z] - 2E[Y]E[Z] \\ &= [E[X^2] - E[X]^2] + [E[Y^2] - E[Y]^2] + [E[Z^2] - E[Z]^2] \\ &\quad + 2[E[XY] - E[X]E[Y]] + 2[E[XZ] - E[X]E[Z]] + 2[E[YZ] - E[Y]E[Z]] \\ &= \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z) + 2\text{Cov}(X, Y) + 2\text{Cov}(X, Z) + 2\text{Cov}(Y, Z) \end{aligned}$$

Under independence, all covariance terms equal zero, giving us:

$$\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z)$$

## Compute each variance term

$$\begin{aligned} \text{Var}_t((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}) &= (\lambda_{i,r} + \lambda_r)^\top \underbrace{\sigma_{g,r} \text{Var}_t(\varepsilon_{t+1}) \sigma_{g,r}^\top}_{\sigma_{g,r} I \sigma_{g,r}^\top = \Sigma_{g,r}} (\lambda_{i,r} + \lambda_r) \\ &= (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r), \\ \text{Var}_t(v_{t+1}) &= \sigma_v^2, \\ \text{Var}_t(\Delta S_{i,t}) &= \text{Var}_t(\Delta(\Gamma_i^\top z_t + u_{i,t})) \quad (\text{finite by assumption}). \end{aligned}$$

## Variance of the global-factor innovation term

We want

$$\text{Var}_t\left((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}\right),$$

where  $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$  is a  $q_r \times 1$  vector of i.i.d. standard normals, and  $\sigma_{g,r}$  is a  $q_r \times q_r$  diagonal matrix with entries  $\sigma_{g1,r}, \dots, \sigma_{gq_r,r}$ .

### Factor out the deterministic coefficients

Since  $(\lambda_{i,r} + \lambda_r)$  and  $\sigma_{g,r}$  are both deterministic conditional on  $\mathcal{F}_t$ ,

$$\text{Var}_t\left((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}\right) = (\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \text{Var}_t(\varepsilon_{t+1}) \sigma_{g,r}^\top (\lambda_{i,r} + \lambda_r).$$

Recall, for any vector  $a$  and random vector  $X$ ,

$$\text{Var}(a^\top X) = a^\top \text{Var}(X) a.$$

### Use the variance of the standard normal vector

By assumption,  $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$ , so

$$\text{Var}_t(\varepsilon_{t+1}) = I.$$

### Simplify the middle term

$$\sigma_{g,r} \text{Var}_t(\varepsilon_{t+1}) \sigma_{g,r}^\top = \sigma_{g,r} I \sigma_{g,r}^\top = \sigma_{g,r} \sigma_{g,r}^\top.$$

### Recognize the definition of $\Sigma_{g,r}$

By the paper's definition,

$$\Sigma_{g,r} := \text{diag}(\sigma_{g1,r}^2, \dots, \sigma_{gq_r,r}^2) = \sigma_{g,r} \sigma_{g,r}^\top.$$

### Substitute back

Therefore,

$$\text{Var}_t\left((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}\right) = (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r).$$

### Combine the pieces

$$\text{Var}_t(Y_{t+1}) = (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta S_{i,t}).$$

Since  $Y_{t+1} | \mathcal{F}_t$  is normal,  $\mathbb{E}_t[e^{Y_{t+1}}] = \exp(\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1})) = 1$ .

$$\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1}) = 0.$$

### Clarafication

The phrase

$$Y_{t+1} | \mathcal{F}_t$$

means that we are considering the conditional distribution of the random variable  $Y_{t+1}$  given the information set (filtration)  $\mathcal{F}_t$ , which contains all information available up to time  $t$ . Saying that

$$Y_{t+1} | \mathcal{F}_t \sim \mathcal{N}(\cdot, \cdot)$$

means that, conditional on what is known at time  $t$ , the distribution of  $Y_{t+1}$  is normal (Gaussian). This assumption follows from the model setup: the shocks  $\varepsilon_{t+1}, v_{t+1}, \Delta S_{i,t}$  are Gaussian and independent, so any linear combination of them, such as  $Y_{t+1}$ , is also Gaussian. Because  $Y_{t+1} | \mathcal{F}_t$  is normal, we can apply the log-normal expectation identity:

$$\mathbb{E}_t[e^{Y_{t+1}}] = \exp(\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1})).$$

## Plugging in the conditional mean and the conditional variance

From above,

$$\mathbb{E}_t[Y_{t+1}] = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r.$$

Remember, we derived

$$\text{Var}_t(Y_{t+1}) = (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_\nu^2 + \text{Var}_t(\Delta S_{i,t}).$$

So,

$$0 = \mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1}) = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \frac{1}{2} \left[ (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_\nu^2 + \text{Var}_t(\Delta S_{i,t}) \right].$$

$$C_t := -r_t - \frac{1}{2}\varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \frac{1}{2} \left[ (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_\nu^2 + \text{Var}_t(\Delta S_{i,t}) \right].$$

$$\mu_{i,r} := [C_t \mid r] = -\tilde{r}_r - \frac{1}{2}\varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \frac{1}{2} \left[ (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_\nu^2 + \text{Var}_t(\Delta(\Gamma_i^\top z_t + u_{i,t})) \right].$$

## Affine form and why the shock $S_{i,t}$ drops out of the mean

We assume the affine decomposition

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + S_{i,t},$$

where  $\mu_{i,r}$  is a constant (intercept),  $\lambda_{i,r}^\top g_t$  is the systematic state term, and  $S_{i,t}$  is an idiosyncratic shock with zero conditional mean:

$$\mathbb{E}[S_{i,t}] = 0.$$

If  $Y = \log(X_{i,t})$ , then

$$\mathbb{E}[e^Y] = \mathbb{E}[X_{i,t}].$$

Taking conditional expectation of the affine form gives

$$\mathbb{E}[X_{i,t}] = \mathbb{E}[\mu_{i,r}] + \mathbb{E}[\lambda_{i,r}^\top g_t] + \mathbb{E}[S_{i,t}] = \mu_{i,r} + \lambda_{i,r}^\top g_t + 0,$$

so the random shock  $S_{i,t}$  drops out and the mean of  $X_{i,t}$  is just the deterministic affine part:

$$\mathbb{E}[X_{i,t}] = \mu_{i,r} + \lambda_{i,r}^\top g_t.$$

That is why, once we factor out the  $g_t$  contribution, the remaining term is simply the constant  $\mu_{i,r}$ .

**noindent**

The USD price of one unit of currency  $i$  delivered at  $t+1$  is  $1/S_{i,t}$ . No-arbitrage implies:

$$1 = \mathbb{E}_t \left[ M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right].$$

Solving under log-normality and stationary fundamentals  $z_t$  yields:

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}, \tag{10}$$

where  $\lambda_{i,r} \in \mathbb{R}^{q_r}$  reflects currency-specific exposure to the  $q_r$  global factors,  $g_t \in \mathbb{R}^{q_r}$ .

## FX Claim Pricing

### Exchange-Rate Convention

We define the exchange rate

$$S_{i,t} = \frac{\text{currency } i}{\text{USD}} \quad (\text{units of currency } i \text{ per USD}).$$

Hence,

$$\frac{1}{S_{i,t}} = \frac{\text{USD}}{\text{currency } i}.$$

### USD Payoff of a Foreign-Currency Claim

A claim that pays  $D_{i,t+1}$  units of currency  $i$  at  $t+1$  has a USD payoff

$$\text{USD payoff at } t+1 = \frac{D_{i,t+1}}{S_{i,t+1}}.$$

### General Pricing with the SDF

By no-arbitrage, the USD price at time  $t$  is

$$P_{i,t} = \mathbb{E}_t \left[ M_{t+1} \frac{D_{i,t+1}}{S_{i,t+1}} \right],$$

where  $M_{t+1}$  is the stochastic discount factor (SDF).

### Special Case: Pure FX Claim

For a pure foreign exchange claim that delivers one unit of currency  $i$ , we set  $D_{i,t+1} = 1$ . Then

$$\text{payoff at } t+1 = \frac{1}{S_{i,t+1}}, \quad P_{i,t} = \mathbb{E}_t \left[ M_{t+1} \frac{1}{S_{i,t+1}} \right].$$

### Replication Argument

Because currency  $i$  is tradeable, one can replicate the claim by paying

$$\frac{1}{S_{i,t}} \text{ USD today to buy one unit of currency } i,$$

and holding it risklessly until  $t+1$ . By no-arbitrage,

$$P_{i,t} = \frac{1}{S_{i,t}}.$$

### Fundamental Euler Equation

Equating the two expressions for  $P_{i,t}$  gives

$$\frac{1}{S_{i,t}} = \mathbb{E}_t \left[ M_{t+1} \frac{1}{S_{i,t+1}} \right] \implies 1 = \mathbb{E}_t \left[ M_{t+1} \frac{S_{i,t}}{S_{i,t+1}} \right].$$

### Log Form

With  $X_{i,t} = \log S_{i,t}$ , we have

$$\frac{S_{i,t}}{S_{i,t+1}} = e^{X_{i,t} - X_{i,t+1}},$$

so the condition can be written as

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}].$$



## Unit Consistency

Note that

$$M_{t+1} \text{ (unitless)} \times \frac{1}{S_{i,t+1}} \left( \frac{\text{USD}}{\text{currency } i} \right) = \text{USD},$$

so  $P_{i,t}$  is expressed in USD.

## Interpretation of Equation (10)

Equation (10) is not merely a declaration; it represents the explicit solution implied by the no-arbitrage condition. The derivation proceeds as follows:

- The no-arbitrage condition can be written as

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}].$$

- Imposing log-normality and assuming stationary fundamentals  $z_t$ , this expectation can be evaluated explicitly using log-linearization techniques in the spirit of Campbell–Shiller.
- The resulting solution yields an affine representation for the log exchange rate:

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}.$$

Accordingly, equation (10) follows directly from the no-arbitrage restriction under log-normality. It is not introduced ad hoc; rather, it is the log-linear solution to the expectation above. The boxed expression for  $\mu_{i,r}$  given in equation (7) simply makes explicit the components contained within the  $\mu_{i,r}$  term in equation (10).

## 5. Regime-Dependent Pass-Through (“Autonomy”)

In order to model how the sensitivity of exchange rates to the global factor vector changes across regimes, we parameterize the factor loadings for currency  $i$  in regime  $r$  as

$$\lambda_{i,r} = (1 - \psi_{i,r}) \odot \bar{\lambda}_i, \quad \bar{\lambda}_i \neq 0 \text{ fixed}, \quad (11)$$

where  $\psi_{i,r} = (\psi_{i1,r}, \dots, \psi_{iq_r,r})' \in [0, 1]^{q_r}$  is the vector of “autonomy” parameters of currency  $i$  from each global factor in regime  $r$ ,  $\bar{\lambda}_i \in \mathbb{R}^{q_r}$  is the baseline (regime-invariant) vector of factor loadings, and  $\odot$  denotes the Hadamard (elementwise) product. A lower  $\psi_{ik,r}$  implies stronger pass-through from factor  $k$  (larger  $|\lambda_{ik,r}|$ ), meaning that the currency’s value is more sensitive to movements in that global factor. A higher  $\psi_{ik,r}$  implies greater autonomy from factor  $k$  in that regime.

The following assumptions describe how volatility and autonomy evolve across regimes, and ensure the statistical tractability of the model:

- A1** For at least one factor  $k$ ,  $\sigma_{g_k, \text{crisis}}^2 > \sigma_{g_k, \text{pre}}^2$  and  $\sigma_{g_k, \text{crisis}}^2 \geq \sigma_{g_k, \text{post}}^2$ . Crisis periods are characterized by heightened volatility in one or more global factors, reflecting increased uncertainty, tighter global financial conditions, or both.
- A2** For all  $i, k$ ,  $\psi_{ik, \text{crisis}} \leq \psi_{ik, \text{pre}}$ . Autonomy compresses during crises as market stress and funding pressures force more synchronized exchange-rate movements with the global factors.
- A3** For a non-empty set  $\mathcal{E}$  (emerging markets) and for at least one factor  $k$ ,  $\psi_{ik, \text{post}} > \psi_{ik, \text{crisis}}$ . Certain EMs regain autonomy from one or more global factors after crises due to policy adjustments, reserve accumulation, or reduced reliance on specific channels of external funding.

## Regime-Dependent Pass-Through and Autonomy

To capture how exchange-rate sensitivity to global factors varies across regimes, the model specifies regime-dependent factor loadings for currency  $i$  in regime  $r$  as

$$\lambda_{i,r} = (1 - \psi_{i,r}) \odot \bar{\lambda}_i, \quad \bar{\lambda}_i \neq 0,$$

where  $\bar{\lambda}_i \in \mathbb{R}^{q_r}$  is the baseline (regime-invariant) vector of factor loadings,  $\psi_{i,r} = (\psi_{i1,r}, \dots, \psi_{iq_r,r})' \in [0, 1]^{q_r}$  is the vector of “autonomy” parameters, and  $\odot$  denotes the Hadamard (elementwise) product.

### Interpretation of Autonomy

Each element  $\psi_{ik,r}$  measures the autonomy of currency  $i$  from global factor  $k$  in regime  $r$ :

- If  $\psi_{ik,r} = 0$ , currency  $i$  has no autonomy from factor  $k$ , and  $\lambda_{ik,r} = \bar{\lambda}_{ik}$ .
- If  $\psi_{ik,r} = 1$ , currency  $i$  achieves full autonomy, with  $\lambda_{ik,r} = 0$ , meaning complete insulation from factor  $k$ .
- Intermediate values  $0 < \psi_{ik,r} < 1$  represent partial autonomy.

Thus, lower  $\psi_{ik,r}$  corresponds to larger  $|\lambda_{ik,r}|$ , implying stronger pass-through of factor  $k$  into currency  $i$ . Higher  $\psi_{ik,r}$  corresponds to weaker pass-through, meaning the exchange rate is less sensitive to global shocks.

### Assumptions Across Regimes

The model imposes the following assumptions to characterize volatility and autonomy in different regimes:

**Crisis Volatility:** For at least one factor  $k$ ,

$$\sigma_{g_k, \text{crisis}}^2 > \sigma_{g_k, \text{pre}}^2, \quad \sigma_{g_k, \text{crisis}}^2 \geq \sigma_{g_k, \text{post}}^2.$$

Crisis regimes are associated with heightened volatility in global factors, reflecting financial stress, uncertainty, and tighter global conditions.

**Autonomy Compression in Crises:** For all  $i, k$ ,

$$\psi_{ik, \text{crisis}} \leq \psi_{ik, \text{pre}}.$$

During crises, market stress forces currencies to move more synchronously with global factors, leading to reduced autonomy.

**Post-Crisis Recovery of Autonomy:** For a nonempty set  $\mathcal{E}$  of emerging markets and at least one factor  $k$ ,

$$\psi_{ik,\text{post}} > \psi_{ik,\text{crisis}}.$$

Certain EMs regain autonomy from global factors post-crisis, often due to policy reforms, reserve accumulation, or reduced dependence on specific external funding channels.

**A4**  $z_t$  and  $u_{i,t}$  are stationary and independent of the global-factor innovations  $\varepsilon_{t+1}$ . Stationarity of  $z_t$  ensures they capture mean-reverting fundamentals without introducing new stochastic trends; stationarity of  $u_{i,t}$  preserves cointegration between  $X_{i,t}$  and its drivers, while independence from  $\varepsilon_{t+1}$  maintains orthogonality between predictable components and shocks.

**Factor-loading structure.** For each regime  $r$ , define the  $N \times q_r$  pass-through matrix

$$\Lambda_r = \begin{bmatrix} \lambda'_{1,r} \\ \vdots \\ \lambda'_{N,r} \end{bmatrix} = \begin{bmatrix} (1 - \psi_{11,r}) \bar{\lambda}_{11} & \dots & (1 - \psi_{1q_r,r}) \bar{\lambda}_{1q_r} \\ \vdots & \ddots & \vdots \\ (1 - \psi_{N1,r}) \bar{\lambda}_{N1} & \dots & (1 - \psi_{Nq_r,r}) \bar{\lambda}_{Nq_r} \end{bmatrix},$$

The prime in  $\lambda'_{i,r} g_t$  denotes the transpose, ensuring that the  $1 \times q_r$  row vector  $\lambda'_{i,r}$  and the  $q_r \times 1$  state vector  $g_t$  are conformable for multiplication, yielding a scalar inner product.

which stacks the currency-specific loading vectors  $\lambda'_{i,r}$  across  $i$ .

Here:

- $\lambda_{i,r} \in \mathbb{R}^{q_r}$  are the currency-specific loadings on the  $q_r$  global factors in regime  $r$ .
- $\psi_{ik,r}$  is the “autonomy” of currency  $i$  from factor  $k$  in regime  $r$ ; lower values imply higher pass-through from that factor.
- $\Gamma_i$  loads on stationary fundamentals  $z_t$ , which do not alter the cointegration rank.
- $u_{i,t}$  is stationary and mean-reverting.

**Exchange-rate system.** With these definitions, the cross-section of exchange rates in regime  $r$  is

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t, \quad (12)$$

where  $\mu_r \in \mathbb{R}^N$  collects the regime-specific intercepts,  $g_t \in \mathbb{R}^{q_r}$  is the vector of nonstationary global factors,  $\Gamma$  is the  $N \times K$  matrix of stationary-factor loadings, and  $u_t$  is the vector of stationary residuals.

## From Individual Equation (10) to System Equation (12)

Equation (10) describes the dynamics of exchange rate  $i$  in regime  $r$ :

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t},$$

where  $\mu_{i,r}$  is the intercept for currency  $i$ ,  $\lambda'_{i,r}$  is the currency-specific loading vector on the global factors  $g_t$ ,  $\Gamma'_i$  loads on the stationary fundamentals  $z_t$ , and  $u_{i,t}$  is the stationary residual.

To move from the individual representation (10) to the system representation (12), we stack all currencies  $i = 1, \dots, N$  into vectors and matrices. Specifically,

- the intercepts  $\mu_{i,r}$  are collected into the vector  $\mu_r \in \mathbb{R}^N$ ;
- the factor loadings  $\lambda'_{i,r}$  are stacked into the  $N \times q_r$  matrix  $\Lambda_r$ ;
- the stationary-factor loadings  $\Gamma'_i$  are stacked into the  $N \times K$  matrix  $\Gamma$ ;
- the residuals  $u_{i,t}$  are collected into the vector  $u_t \in \mathbb{R}^N$ ;
- the dependent variables  $X_{i,t}$  are collected into the vector  $X_t \in \mathbb{R}^N$ .

In this way, the  $N$  individual currency equations in (10) can be written compactly as the system

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

which is equation (12). This representation makes clear that (12) is simply the vectorized form of (10), capturing the cross-section of exchange rates in regime  $r$ .

## 4. Cointegration Rank and Regime Dependence

Given the affine representation

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

the long-run dimension of comovement among the  $N$  exchange rates in regime  $r$  is determined by the rank of the pass-through matrix  $\Lambda_r$ .

The vector  $g_t \in \mathbb{R}^{q_r}$  collects the  $q_r$  nonstationary global factors relevant in regime  $r$ , while  $\Lambda_r \in \mathbb{R}^{N \times q_r}$  stacks the currency-specific loading vectors  $\lambda'_{i,r}$ . Stationary fundamentals  $z_t$  and residuals  $u_t$  do not affect the cointegration rank because they are  $I(0)$ .

The number of distinct common stochastic trends transmitted to  $X_t$  in regime  $r$  is

$$\text{rank}(\Lambda_r) \leq \min(N, q_r).$$

This yields the standard cointegration rank result:

- If  $\text{rank}(\Lambda_r) = 1$ , there is a *single* nonstationary common component, and the system has  $N - 1$  cointegrating relations (the single-factor baseline).
- If  $\text{rank}(\Lambda_r) = k > 1$ , there are  $N - k$  cointegrating relations (multiple independent global trends drive the system).
- If  $\text{rank}(\Lambda_r) = N$ , there is no cointegration (each currency contains its own stochastic trend).

**Regime variation.** Both the factor volatilities  $\sigma_{g_k,r}^2$  and the autonomy parameters  $\psi_{ik,r}$  can vary across regimes, altering the effective rank of  $\Lambda_r$ . For example:

- A crisis may *increase* factor volatilities (making existing common trends more dominant) and/or *compress* autonomy parameters (aligning currencies more closely with the factors).
- If a new global factor emerges—e.g., a pandemic-specific risk driver during COVID—it can raise  $\text{rank}(\Lambda_r)$  and reduce the number of cointegrating relations. In the extreme, if enough independent trends are introduced, cointegration may disappear even if it was present pre-crisis.

## Exchange Rates, Global Factors, and Cointegration

We model the vector of exchange rates  $X_t$  in regime  $r$  as

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

where

- $X_t$  is the  $N \times 1$  vector of exchange rates,
- $g_t$  is a  $q_r \times 1$  vector of nonstationary global factors (integrated of order one),
- $\Lambda_r$  is the  $N \times q_r$  loading matrix linking global factors to exchange rates,
- $z_t$  are stationary fundamentals with loadings  $\Gamma$ , and
- $u_t$  is an idiosyncratic stationary component.

## Role of $\Lambda_r$ and Cointegration Rank

Only the term  $\Lambda_r g_t$  contributes to the nonstationary part of  $X_t$ . The number of independent global stochastic trends depends on the rank of  $\Lambda_r$ :

$$\text{rank}(\Lambda_r) = k \implies \text{Number of cointegrating relations} = N - k.$$

Examples:

- If  $\text{rank}(\Lambda_r) = 1$ , all exchange rates are driven by a single global trend, so there are  $N - 1$  cointegrating relations.
- If  $\text{rank}(\Lambda_r) = k > 1$ , multiple independent global forces drive the system, and cointegration is weaker.
- If  $\text{rank}(\Lambda_r) = N$ , each currency has its own independent trend, implying no cointegration at all.

## Regime Dependence

The rank of  $\Lambda_r$  may vary across regimes due to changes in volatility or structural shifts:

- In crises, global factor volatilities  $\sigma_{g_{k,r}}^2$  often increase and country autonomy parameters  $\psi_{ik,r}$  shrink. This compresses exchange rate movements into fewer common directions, effectively reducing  $\text{rank}(\Lambda_r)$  and increasing cointegration.
- Conversely, when new global factors emerge (e.g. COVID-19 shock), the effective dimension of  $g_t$  may rise. Then  $\text{rank}(\Lambda_r)$  increases, reducing the number of cointegrating relations, and in extreme cases eliminating cointegration.

## Intuition

The matrix  $\Lambda_r$  describes how exchange rates are tied to underlying global forces:

- A single dominant force  $\implies$  tight co-movement and strong cointegration.
- Several independent forces  $\implies$  weaker long-run ties.
- Each currency tied to its own force  $\implies$  no cointegration at all.

Thus, the degree of long-run linkage among currencies is not constant but regime-dependent, shifting with global volatility and structural shocks.

**Proposition 1 (Rank–Cointegration Link).** Given regime  $r$  and assumptions A1–A4, the cointegration rank of  $X_t$  is

**$N = \text{number of currencies}$**

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r).$$

*Economic interpretation:* This result formalizes the direct link between the number of independent nonstationary global trends and the strength of long-run exchange-rate linkages. If  $\text{rank}(\Lambda_r) = 1$ , all persistent variation is driven by a single global factor (e.g., the USD funding/liquidity factor), yielding  $N - 1$  independent stationary spreads. As  $\text{rank}(\Lambda_r)$  increases, additional nonstationary drivers—such as commodity-price trends, regional funding shocks, or segmented risk premia—introduce new sources of long-run divergence, reducing the number of cointegrating relations. In the extreme case  $\text{rank}(\Lambda_r) = N$ , each currency has its own independent trend and no long-run stationary spread survives.

## Proposition 1 Explained

The cointegration rank of  $X_t$  is given by

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r),$$

where  $N$  is the number of time series and  $\Lambda_r$  is the loading matrix in regime  $r$ .

The rank of  $\Lambda_r$  measures the number of **independent nonstationary global trends** that drive the system. Subtracting this number from  $N$  gives the **number of independent stationary relations** (cointegrating vectors). In other words:

$$\text{Number of stationary spreads} = N - (\text{Number of global stochastic trends}).$$

**Case 1:**  $\text{rank}(\Lambda_r) = 1$ . All persistent variation is driven by a single global factor (e.g., a USD funding/liquidity factor).

$$\text{Cointegration rank} = N - 1.$$

This implies there are  $N - 1$  independent stationary spreads that eliminate the single common trend.

**Case 2:**  $\text{rank}(\Lambda_r) = k$ . There are  $k$  distinct nonstationary global drivers (e.g., commodity price trends, regional funding shocks, segmented risk premia).

$$\text{Cointegration rank} = N - k.$$

Each additional global driver introduces more long-run divergence and reduces the number of stationary relations.

**Extreme case:**  $\text{rank}(\Lambda_r) = N$ . Each series has its own independent trend.

$$\text{Cointegration rank} = 0.$$

In this case, no stationary long-run spreads survive.

**Proposition 2 (Crisis Co-Movement Amplification).** If  $\text{rank}(\Lambda_r)$  remains unchanged in crisis, but (i)  $\sigma_{g_{k,r}}^2$  rises for one or more factors, and (ii)  $|\lambda_{ik,r}|$  converge across currencies (autonomy compression), then long-run correlations increase.

*Economic interpretation:* Crises amplify co-movement through two reinforcing channels:

1. **Variance channel:** Higher volatility in the common factors increases their share of total long-horizon variance.

2. **Loading-alignment channel:** Lower dispersion in  $\psi_{ik,r}$  makes currency responses more similar in magnitude and sign, reducing idiosyncratic offsets.

Even without introducing new stochastic trends, these channels produce tighter long-run alignment—a pattern consistent with the “risk-on/risk-off” behavior observed during global financial stress.

## Proposition 2 Explained

Proposition 2 states that during a crisis episode, **even though the number of common stochastic trends (the cointegration rank of  $\Lambda_r$ ) does not change**, exchange rates exhibit *tighter long-run co-movement*. This arises through two distinct but reinforcing channels: increased factor variance and stronger alignment of factor loadings.

### Economic Interpretation

Crises amplify co-movement between exchange rates through two reinforcing channels. The key point is that even without introducing new stochastic trends, changes in variance and factor loadings make existing common shocks more dominant, forcing currencies to move together more closely.

### Variance Channel (Dominance of Common Shocks):

**Cause:** In crises, volatility of the global driving factors (e.g., funding liquidity, risk appetite, commodity shocks) rises sharply.

**Mechanism:** When  $\sigma_{g_{k,r}}^2$  increases, these global shocks account for a much larger fraction of the total variance of exchange rates. The contribution of idiosyncratic country-specific shocks becomes relatively smaller.

**Effect:** Since most of the variation across currencies is now explained by the same set of highly volatile factors, correlations across exchange rates mechanically increase. Put simply, higher volatility in global factors forces currencies to co-move more tightly.

Each common global factor evolves as

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_{k,r}} \varepsilon_{k,t+1},$$

with factor volatility  $\sigma_{g_{k,r}}^2$  in regime  $r$ . Collecting all factors, we have covariance matrix

$$\Sigma_{g,r} = \text{diag}(\sigma_{g_{1,r}}^2, \dots, \sigma_{g_{q_r,r}}^2).$$

Equation (10) decomposes the excess return of currency  $i$  at time  $t$  as

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t},$$

where:

- $X_{i,t}$  is the excess return (in logs) of currency  $i$  at time  $t$ .
- $\mu_{i,r}$  is a currency-specific constant mean term within region  $r$ .
- $g_t \in \mathbb{R}^{q_r}$  is the vector of  $q_r$  global risk factors, and  $\lambda_{i,r} \in \mathbb{R}^{q_r}$  is the corresponding factor loading vector that measures currency  $i$ 's exposure to these global shocks. The term  $\lambda'_{i,r} g_t$  therefore captures the component of returns driven by common global forces.
- $z_t$  is the vector of stationary fundamentals (e.g. interest differentials, inflation measures, or macroeconomic indicators), and  $\Gamma_i$  is the vector of sensitivities of currency  $i$  to these fundamentals. The contribution of fundamentals is given by  $\Gamma'_i z_t$ .
- $u_{i,t}$  is an idiosyncratic shock, unique to currency  $i$ , not explained by common factors or fundamentals.

Thus, the decomposition shows that the return on currency  $i$  can be split into a mean term, a systematic component explained by global factors, a component explained by local fundamentals, and an idiosyncratic noise term. The variance of this component is

$$\text{Var}[\lambda'_{i,r} g_t] = \lambda'_{i,r} \Sigma_{g,r} \lambda_{i,r}.$$



If  $\sigma_{g_{k,r}}^2$  rises in crisis, then  $\Sigma_{g,r}$  inflates. This increases the contribution of systematic shocks relative to idiosyncratic variance, meaning that factor-driven co-movement accounts for a larger share of long-run fluctuations. In other words, factor variance dominates, making different currencies move together more strongly.

### Loading-Alignment Channel (Synchronization of Responses):

**Cause:** During crises, the dispersion in factor loadings  $\psi_{ik,r}$  falls. In other words, different currencies respond in a more similar way to the same common shocks.

**Mechanism:** Normally, currencies may load differently on global factors: some more strongly, some weakly, and some even with opposite signs. This creates offsets that reduce measured correlation. In crises, loadings converge in both magnitude and sign, compressing these differences.

**Effect:** Because responses to global shocks become more uniform, idiosyncratic offsets vanish and currency movements align more closely. This structural compression increases the degree of co-movement across currencies.

Each exchange rate  $X_{i,t}$  is represented as

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t},$$

where  $\lambda_{i,r}$  are the factor loadings,  $z_t$  are stationary controls, and  $u_{i,t}$  is idiosyncratic stationary noise. Suppose two currencies  $i$  and  $j$  load on the same factor  $g_t$ . Their long-horizon correlation is approximately

$$\rho_{ij} = \frac{\lambda'_{i,r} \Sigma_{g,r} \lambda_{j,r}}{\sqrt{(\lambda'_{i,r} \Sigma_{g,r} \lambda_{i,r}) (\lambda'_{j,r} \Sigma_{g,r} \lambda_{j,r})}}.$$

If the loadings  $\lambda_{i,r}$  and  $\lambda_{j,r}$  become more aligned (their direction vectors approach each other), then the numerator approaches the denominator, and  $\rho_{ij} \rightarrow 1$ . This is the “autonomy compression” channel: currencies lose heterogeneity in their response to global shocks, and their dynamics become more synchronized.

### Derivation

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}.$$

We start from the factor representation:

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}.$$

Here,  $g_t \in \mathbb{R}^{q_r}$  are global factors with covariance matrix  $\Sigma_{g,r} = \text{Var}(g_t)$ , and  $u_{i,t}$  are idiosyncratic shocks that are independent across assets and uncorrelated with  $g_t$ . Since the idiosyncratic terms drop out when computing cross-covariances, we focus only on the factor-driven part:

$$\text{Cov}(X_{i,t}, X_{j,t}) = \text{Cov}(\lambda'_{i,r} g_t, \lambda'_{j,r} g_t).$$

By the bilinearity of covariance,

$$\text{Cov}(\lambda'_{i,r} g_t, \lambda'_{j,r} g_t) = \lambda'_{i,r} \text{Var}(g_t) \lambda_{j,r}.$$

Recall that for any random vectors  $a, b$  and constant vectors  $c, d$ :

$$\text{Cov}(c'a, d'b) = c' \text{Cov}(a, b) d.$$

In our case we take  $a = b = g_t$ , with  $\text{Cov}(g_t, g_t) = \Sigma_{g,r}$ , and  $c = \lambda_{i,r}$ ,  $d = \lambda_{j,r}$ . Thus,

$$\text{Cov}(\lambda'_{i,r} g_t, \lambda'_{j,r} g_t) = \lambda'_{i,r} \Sigma_{g,r} \lambda_{j,r}.$$

Since  $\text{Var}(g_t) = \Sigma_{g,r}$ , this becomes

$$\text{Cov}(X_{i,t}, X_{j,t}) = \lambda'_{i,r} \Sigma_{g,r} \lambda_{j,r}.$$

Next, the variance of  $X_{i,t}$  is

$$\text{Var}(X_{i,t}) = \text{Var}(\lambda'_{i,r} g_t) = \lambda'_{i,r} \Sigma_{g,r} \lambda_{i,r}.$$

$$X_{i,t} = \lambda'_{i,r} g_t = \sum_{k=1}^{q_r} \lambda_{i,r,k} g_{t,k}.$$

Because variance involves all pairwise covariance terms,

$$\begin{aligned} \text{Var}(X_{i,t}) &= \text{Var}\left(\sum_{k=1}^{q_r} \lambda_{i,r,k} g_{t,k}\right) = \sum_{k=1}^{q_r} \sum_{\ell=1}^{q_r} \lambda_{i,r,k} \lambda_{i,r,\ell} \text{Cov}(g_{t,k}, g_{t,\ell}). \\ &= \lambda'_{i,r} \Sigma_{g,r} \lambda_{i,r}. \end{aligned}$$

Similarly, the variance of  $X_{j,t}$  is

$$\text{Var}(X_{j,t}) = \lambda'_{j,r} \Sigma_{g,r} \lambda_{j,r}.$$

Finally, by definition of correlation,

$$\rho_{ij} = \frac{\text{Cov}(X_{i,t}, X_{j,t})}{\sqrt{\text{Var}(X_{i,t}) \text{Var}(X_{j,t})}} = \frac{\lambda'_{i,r} \Sigma_{g,r} \lambda_{j,r}}{\sqrt{(\lambda'_{i,r} \Sigma_{g,r} \lambda_{i,r})(\lambda'_{j,r} \Sigma_{g,r} \lambda_{j,r})}}.$$

## Summary

Thus, there are two reinforcing drivers of crisis co-movement:

- Increased factor volatility ( $\Sigma_{g,r}$  larger) boosts the relative weight of common factors versus idiosyncratic shocks, magnifying the importance of global drivers.
- Stronger alignment of loadings ( $\lambda_{i,r}$  and  $\lambda_{j,r}$  becoming more similar) pushes pairwise correlations toward one, tightening co-movement.

**Proposition 3 (Rank Change in Crisis).** If new persistent global factors emerge or autonomy shifts such that  $\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}})$ , the number of cointegrating relations falls. If  $\text{rank}(\Lambda_{\text{crisis}}) = N$ , cointegration disappears entirely.

*Economic interpretation:* Certain crises—such as the COVID-19 shock—introduce structural changes that are not spanned by pre-crisis factors: e.g., pandemic-specific risk premia, supply-chain dislocations, or regime-specific commodity shocks. If these shocks load persistently across currencies but in a pattern independent of pre-existing factors, they increase the effective dimension of  $g_t$  and the rank of  $\Lambda_r$ . The result is fewer cointegrating relations, with the extreme of rank  $N$  corresponding to a complete breakdown of long-run linkage.

## Proposition 3 Explained

Let  $\Lambda_{\text{pre}}$ : factor loadings matrix before the crisis,  $\Lambda_{\text{crisis}}$ : factor loadings matrix during the crisis,  $N$ : number of currencies/assets in the system.

### Proposition

If new persistent global factors emerge (or existing ones shift) such that

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}),$$

then the number of cointegrating relations falls. If

$$\text{rank}(\Lambda_{\text{crisis}}) = N,$$

then *cointegration disappears entirely*.

### Intuition

**Cointegration** relies on there being fewer independent driving forces (factors) than there are assets; this creates long-run equilibrium relations (some linear combinations cancel common factors). If a crisis introduces *new independent shocks* not spanned by pre-crisis factors, the effective dimensionality increases.

Result: fewer linear constraints  $\Rightarrow$  fewer cointegrating relations.

In the extreme, if the number of independent driving forces equals the number of assets ( $\text{rank} = N$ ), there is no redundancy left  $\Rightarrow$  **no cointegration**.

### Economic Interpretation

**Before crisis:** exchange rates share common global factors (e.g., interest-rate differentials, commodity prices, monetary policy), generating long-run linkages.

**During crisis:** new shocks appear that were previously irrelevant (pandemic-specific risk premia, supply-chain breakdowns, regime- or sector-specific commodity shocks). These shocks load *independently* across currencies:

$$\text{dimension of } g_t \text{ increases} \Rightarrow \text{rank}(\Lambda) \uparrow \Rightarrow \text{fewer cointegrating relations}$$

**Extreme case:** if every currency has its own unique persistent driver ( $\text{rank} = N$ ), long-run linkages collapse; currencies drift without stable equilibrium relations.

### Regime Dependence: When Crises *Reduce* vs. *Increase* Rank

Crises can operate through two distinct mechanisms. Recognizing which mechanism dominates resolves any apparent contradiction:

- **Volatility-dominated (amplification) crises:** global factor volatilities surge and country-specific autonomy shrinks. Movements compress into *fewer* common directions  $\Rightarrow \text{rank}(\Lambda)$  *decreases*  $\Rightarrow$  cointegration *increases*.
- **Structural-shift (new-factor) crises:** genuinely new, persistent global forces emerge (e.g., COVID-19). The effective dimension of  $g_t$  rises  $\Rightarrow \text{rank}(\Lambda)$  *increases*  $\Rightarrow$  cointegration *decreases*.

**Proposition 2 doesn't even talk about cointegration.**

**It is talking about correlation. Ignore the whole volatility dominated bullet point. It doesn't make sense.**

**Proposition 4 (Post-Crisis Divergence).** If  $\text{rank}(\Lambda_{\text{post}}) < \text{rank}(\Lambda_{\text{crisis}})$  and the dispersion in autonomy parameters  $\psi_{ik,r}$  widens, long-run correlations weaken.

*Economic interpretation:* The dissipation of certain crisis-specific global factors in the post-crisis period reduces  $\text{rank}(\Lambda_r)$ , potentially restoring some cointegrating relations. However, as countries pursue differentiated policy responses, alter reserve portfolios, or shift trade and financial linkages, heterogeneity in autonomy parameters may increase. This heterogeneity reduces the absolute correlation of long-horizon returns, producing a “partial decoupling” effect in which some pre-crisis integration is permanently lost.

## Proposition 4 Explained

Let  $\Lambda_{\text{crisis}}$ : factor loadings matrix during the crisis,  $\Lambda_{\text{post}}$ : factor loadings matrix in the post-crisis period.

### Proposition

If

$$\text{rank}(\Lambda_{\text{post}}) < \text{rank}(\Lambda_{\text{crisis}})$$

and the dispersion in country autonomy parameters  $\psi_{ik,r}$  widens, then long-run correlations weaken.

### Intuition

During crisis, new global shocks appear, raising  $\text{rank}(\Lambda_{\text{crisis}})$  and breaking cointegration (Proposition 3). After crisis, some of those crisis-specific shocks dissipate, so  $\text{rank}(\Lambda_{\text{post}})$  falls. In principle, this reduction in rank allows certain cointegrating relations to be restored. However, post-crisis, countries often diverge in their behavior: different monetary and fiscal policies, altered reserve portfolio strategies, and shifts in trade and financial linkages. This heterogeneity appears in the dispersion of autonomy parameters  $\psi_{ik,r}$ . The wider the dispersion, the less tightly returns move together across currencies. Result, even though the number of factors shrinks (rank falls), long-run correlations weaken rather than fully recover.

### Economic Interpretation

Dissipation of crisis-specific shocks reduces rank, which *could* restore some long-run equilibrium linkages. But in practice, post-crisis divergence in policy and structural choices across countries produces weaker correlations of long-horizon returns. The outcome is a partial decoupling effect: some pre-crisis integration is permanently lost, even though the shock that created the crisis has faded.

## 10 Appendix: Complete Proofs

### Proof of Proposition 1 (Cointegration rank $N - 1$ )

For any  $b \in \mathbb{R}^N$ , note:

$$b'X_t = b'\mu_r + (b'\Lambda_r)g_t + b'\Gamma z_t + b'u_t.$$

Since  $z_t$  and  $u_t$  are stationary,  $b'\Gamma z_t + b'u_t$  is stationary  $I(0)$ . The only potentially nonstationary component is  $(b'\Lambda_r)g_t$ , because  $g_t$  is  $I(1)$  and its components are assumed not to be cointegrated (so they represent independent stochastic trends). Thus:

- If  $b'\Lambda_r = 0$ , then  $b'X_t$  is a constant plus stationary variables  $I(0)$  terms  $\Rightarrow I(0)$ .
- If  $b'\Lambda_r \neq 0$ , then  $b'X_t$  inherits an  $I(1)$  component  $\Rightarrow I(1)$ .

Hence, the cointegrating space is  $\{b \in \mathbb{R}^N : b'\Lambda_r = 0\}$ , of dimension  $N - 1$  when  $q_r = 1$  and  $\Lambda_r \neq 0$ .

## What Does Cointegration Mean?

### Definition

Two or more time series are said to be *cointegrated* if:

- Each series is individually nonstationary (typically integrated of order one,  $I(1)$ , i.e. they behave like random walks).
- Some linear combination of these series is stationary ( $I(0)$ ), meaning it reverts to a stable mean over time.

### Simple Example

Suppose we have two exchange rates, both of which are  $I(1)$ :

$$X_t \sim I(1), \quad Y_t \sim I(1).$$

Individually, each process drifts without bound. However, if their difference

$$Z_t = X_t - Y_t$$

is stationary ( $I(0)$ ), then the spread between them is mean-reverting and does not drift away indefinitely. In this case,  $X_t$  and  $Y_t$  are cointegrated with cointegrating vector  $(1, -1)$ .

### Interpretation in FX Markets

In the context of exchange rates, cointegration means that although each log exchange rate  $X_{i,t}$  is nonstationary due to global shocks, there exist long-run equilibrium relationships among currencies. Certain linear combinations of exchange rates “cancel out” the common stochastic trends, yielding stable spreads.

- If all  $N$  exchange rates are driven by a single global USD factor, then there exist  $N - 1$  independent stationary spreads between them.

## Economic Intuition

Cointegration captures the idea that even though each exchange rate drifts like a random walk, certain relative values between them are anchored by economic forces and do not diverge without bound. In this paper's results:

- During the **GFC**, cointegration strengthened: currencies moved more closely together and relative spreads were tighter.
- During **COVID**, cointegration weakened: spreads broke down and currencies exhibited more independence.

## Proof of Proposition 1 (Cointegration Rank = $N - \text{rank}(\Lambda_r)$ )

We now provide a step-by-step explanation of why the cointegration rank of the system is determined by the rank of the factor-loading matrix  $\Lambda_r$ .

### Setup

We consider the model

$$X_t = \mu + \Lambda_r g_t + \nu_t,$$

where:

- $X_t \in \mathbb{R}^N$ : vector of exchange rates,
- $\Lambda_r \in \mathbb{R}^{N \times K}$ : factor loading matrix,
- $g_t \in \mathbb{R}^K$ : global stochastic trends, each integrated of order one ( $I(1)$ ),
- $\nu_t$ : stationary noise.

Thus, the *nonstationary component* of  $X_t$  is entirely captured by  $\Lambda_r g_t$ .

### Idea of Cointegration

A linear combination  $\beta^\top X_t$  is stationary if the nonstationary parts cancel. Substituting:

$$\beta^\top X_t = \beta^\top \mu + \beta^\top \Lambda_r g_t + \beta^\top \nu_t.$$

- $\beta^\top \mu$  is constant (hence stationary).
- $\beta^\top \nu_t$  is stationary by assumption.
- The only potentially nonstationary term is  $\beta^\top \Lambda_r g_t$ , since  $g_t \sim I(1)$ .

Therefore, the condition for cointegration is:

$$\beta^\top \Lambda_r g_t \text{ is stationary.}$$

But because  $g_t$  is  $I(1)$ , this holds *if and only if*

$$\beta^\top \Lambda_r = 0.$$

### Null Space Argument

The set of vectors  $\beta$  satisfying  $\beta^\top \Lambda_r = 0$  is precisely the *left null space* of  $\Lambda_r$ . The dimension of this null space is

$$\dim(\{\beta \in \mathbb{R}^N : \beta^\top \Lambda_r = 0\}) = N - \text{rank}(\Lambda_r).$$

This dimension equals the number of linearly independent cointegrating vectors.

## Intuition

- $\Lambda_r$  determines how many independent nonstationary shocks actually drive  $X_t$ .
- If  $\Lambda_r$  has rank  $r$ , then there are  $r$  independent nonstationary directions.
- This leaves  $N - r$  independent stationary linear combinations, i.e. the cointegrating relations.

In plain words: the *cointegration rank* is the number of exchange-rate relationships that eliminate the common stochastic trends. It is given by the dimension of the null space of the factor-loading matrix  $\Lambda_r$ .

### (Cointegration rank $N - \text{rank}(\Lambda_r)$ )

Given regime  $r$  and assumptions A1–A4, the cointegration rank of  $X_t$  is:

$$\text{Cointegration rank of } X_t = N - \text{rank}(\Lambda_r).$$

*Proof.* Consider the long-run factor structure:

$$X_t = \mu + \Lambda_r g_t + \nu_t,$$

where:

- $X_t \in \mathbb{R}^N$  is a vector of exchange rates (e.g., log bilateral exchange rates),
- $\Lambda_r \in \mathbb{R}^{N \times K}$  is the matrix of factor loadings,
- $g_t \in \mathbb{R}^K$  is a vector of global stochastic trends, assumed to be integrated of order one ( $I(1)$ ),
- $\nu_t$  is a stationary vector process.

By assumption A2, the components of  $g_t$  are mutually uncorrelated and not cointegrated. Thus,  $g_t$  represents  $K$  independent stochastic trends. The nonstationarity in  $X_t$  arises only from the term  $\Lambda_r g_t$ . Let  $r := \text{rank}(\Lambda_r) \leq \min(N, K)$ . Then the image of  $g_t$  under  $\Lambda_r$  spans an  $r$ -dimensional subspace of  $\mathbb{R}^N$ . This implies that the nonstationary component of  $X_t$  lies entirely within an  $r$ -dimensional subspace. Let us now consider the existence of cointegrating vectors. A cointegrating vector  $\beta \in \mathbb{R}^N$  is such that the linear combination  $\beta^\top X_t$  is stationary:

$$\beta^\top X_t = \beta^\top \mu + \beta^\top \Lambda_r g_t + \beta^\top \nu_t.$$

Since  $\mu$  is constant and  $\nu_t$  is stationary, the stationarity of  $\beta^\top X_t$  depends entirely on whether  $\beta^\top \Lambda_r g_t$  is stationary. But  $g_t$  is  $I(1)$ , so  $\beta^\top \Lambda_r g_t$  is stationary if and only if:

$$\beta^\top \Lambda_r = 0.$$

That is, cointegrating vectors lie in the left null space of  $\Lambda_r$ . The dimension of this null space is:

$$\dim(\{\beta \in \mathbb{R}^N : \beta^\top \Lambda_r = 0\}) = N - \text{rank}(\Lambda_r).$$

Hence, the number of linearly independent cointegrating vectors—i.e., the cointegration rank of  $X_t$ —is:

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r).$$

## Proposition 2 (Crisis Co-Movement Amplification of Long Horizon Currency Returns)

We begin with the factor model for currency  $i$  in regime  $r$ :

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top z_t + u_{i,t},$$

where  $g_t$  are global stochastic trends (e.g., integrated),  $z_t$  are stationary domestic/regional factors, and  $u_{i,t}$  is idiosyncratic noise.

Taking first differences:

$$\Delta X_{i,t} = \lambda_{i,r}^\top \Delta g_t + \Delta(\Gamma_i^\top z_t + u_{i,t}).$$

Define the cumulative return over horizon  $H$  as:

$$Y_{i,H} \equiv \sum_{h=1}^H \Delta X_{i,t+h} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \varepsilon_{i,H},$$

where  $\varepsilon_{i,H} = \sum_{h=1}^H \Delta(\Gamma_i^\top z_{t+h} + u_{i,t+h})$  is a sum of stationary terms.



### Variance of the long-run return

Because  $g_t$  is a vector of random walks,

$$\text{Var}(g_{t+H} - g_t) = H\Sigma_{g,r},$$

and we assume  $\varepsilon_{i,H}$  is uncorrelated with  $g_t$ . Then:

$$\text{Var}(Y_{i,H}) \approx H\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + \text{Var}(\varepsilon_{i,H}).$$

Since  $\varepsilon_{i,H}$  is a sum of stationary shocks:

$$\text{Var}(\varepsilon_{i,H}) \approx H\sigma_{\varepsilon,i}^2.$$

### Factor share of total variance

The share of total variance explained by global factors is:

$$\frac{\text{Factor variance}}{\text{Total variance}} = \frac{H\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r}}{H\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + H\sigma_{\varepsilon,i}^2} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r}}{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + \sigma_{\varepsilon,i}^2}.$$

As  $\Sigma_{g,r}$  increases, this ratio increases. Thus, the common global factors explain a larger fraction of the long-run variance. This is the *variance channel* of co-movement amplification.

### Correlation between long-run returns

For any pair  $i, j$ , define:

$$\rho_{ij}^{(H)} = \frac{\text{Cov}(Y_{i,H}, Y_{j,H})}{\sqrt{\text{Var}(Y_{i,H}) \text{Var}(Y_{j,H})}}.$$

We have:

$$\begin{aligned} \text{Cov}(Y_{i,H}, Y_{j,H}) &= H\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + \text{Cov}(\varepsilon_{i,H}, \varepsilon_{j,H}), \\ \text{Var}(Y_{i,H}) &= H\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + H\sigma_{\varepsilon,i}^2, \quad \text{Var}(Y_{j,H}) = H\lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + H\sigma_{\varepsilon,j}^2. \end{aligned}$$

Divide numerator and denominator by  $H$ :

$$\rho_{ij}^{(H)} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + \frac{1}{H} \text{Cov}(\varepsilon_{i,H}, \varepsilon_{j,H})}{\sqrt{(\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + \frac{1}{H} \sigma_{\varepsilon,i}^2) (\lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + \frac{1}{H} \sigma_{\varepsilon,j}^2)}}.$$

As  $H \rightarrow \infty$ , the idiosyncratic components vanish:

$$\lim_{H \rightarrow \infty} \rho_{ij}^{(H)} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r}}{\sqrt{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} \cdot \lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r}}}.$$

This is the cosine of the angle between the vectors  $\Sigma_{g,r}^{1/2} \lambda_{i,r}$  and  $\Sigma_{g,r}^{1/2} \lambda_{j,r}$ .

If the factor loadings are aligned (i.e.,  $\lambda_{i,r} \approx \lambda_{j,r}$ ), then:

$$\lim_{H \rightarrow \infty} \rho_{ij}^{(H)} \rightarrow 1.$$

Thus, greater similarity in how currencies respond to global shocks leads to stronger long-run correlation. This is the *loading-alignment channel*.

## Conclusion

During crises, increases in global factor volatility and convergence in cross-currency loadings jointly amplify long-run return correlations and the dominance of common shocks. This explains tighter global alignment in currency returns during episodes of global financial stress. ■

## Proposition 2 (Crisis Co-Movement Amplification of Long-Horizon Currency Returns)

### Setup and Assumptions

For currency  $i$  in regime  $r$ ,

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top z_t + u_{i,t},$$

### Horizon- $H$ Return and Telescoping of Stationary Parts

First-difference the model:

$$\Delta X_{i,t} = \lambda_{i,r}^\top \Delta g_t + \Delta(\Gamma_i^\top z_t + u_{i,t}).$$

### Deriving the First-Difference Equation

Fix a regime  $r$  in which the loadings are constant over time. Start from the level model

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top z_t + u_{i,t}, \quad X_{i,t-1} = \mu_{i,r} + \lambda_{i,r}^\top g_{t-1} + \Gamma_i^\top z_{t-1} + u_{i,t-1}.$$

Define  $\Delta X_{i,t} \equiv X_{i,t} - X_{i,t-1}$  and subtract:

$$\begin{aligned}\Delta X_{i,t} &= (\mu_{i,r} - \mu_{i,r}) + \lambda_{i,r}^\top (g_t - g_{t-1}) + \Gamma_i^\top (z_t - z_{t-1}) + (u_{i,t} - u_{i,t-1}) \\ &= \lambda_{i,r}^\top \Delta g_t + \Gamma_i^\top \Delta z_t + \Delta u_{i,t} \\ &= \lambda_{i,r}^\top \Delta g_t + \Delta(\Gamma_i^\top z_t + u_{i,t}),\end{aligned}$$

where the last equality uses that  $\Gamma_i$  is time-invariant within regime  $r$ , so  $\Delta(\Gamma_i^\top z_t) = \Gamma_i^\top \Delta z_t$ . Define the  $H$ -period (cumulative) return

$$Y_{i,H} \equiv \sum_{h=1}^H \Delta X_{i,t+h} = \lambda_{i,r}^\top \sum_{h=1}^H \Delta g_{t+h} + \sum_{h=1}^H \Delta(\Gamma_i^\top z_{t+h} + u_{i,t+h}).$$

Using the telescoping identity  $\sum_{h=1}^H \Delta x_{t+h} = x_{t+H} - x_t$ ,

$$Y_{i,H} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \underbrace{\Gamma_i^\top (z_{t+H} - z_t) + (u_{i,t+H} - u_{i,t})}_{\varepsilon_{i,H}}.$$

## H-step Increment with Drift and Scaled Shocks

For each factor  $k$  in regime  $r$ ,

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_{k,r}} \varepsilon_{k,t+1}, \quad \mathbb{E}[\varepsilon_{k,t+1}] = 0, \quad \text{Var}(\varepsilon_{k,t+1}) = 1.$$

Iterating  $H$  steps and telescoping gives

$$g_{k,t+H} - g_{k,t} = \sum_{h=1}^H (g_{k,t+h} - g_{k,t+h-1}) = H \mu_{k,r} + \sigma_{g_{k,r}} \sum_{h=1}^H \varepsilon_{k,t+h}.$$

## Vector Form

Let  $\mu_r = (\mu_{1,r}, \dots, \mu_{q_r,r})^\top$  and  $L_r = \text{diag}(\sigma_{g_{1,r}}, \dots, \sigma_{g_{q_r,r}})$ . Then

$$g_{t+H} - g_t = H \mu_r + L_r \sum_{h=1}^H \varepsilon_{t+h} = \sum_{h=1}^H (\mu_r + L_r \varepsilon_{t+h})$$

we will show that

$$\mathbb{E}[g_{t+H} - g_t] = H \mu_r, \quad \text{Var}(g_{t+H} - g_t) = H \Sigma_{g,r}.$$

## Variance Scaling (“Variance Channel”)

We have

$$\text{Var}(g_{t+H} - g_t) = H \Sigma_{g,r}.$$

### Derivation of $\text{Var}(g_{t+H} - g_t) = H \Sigma_{g,r}$

Start from the stated factor law in regime  $r$ :

$$g_{t+1} = g_t + \mu_r + L_r \varepsilon_{t+1}, \quad \mathbb{E}[\varepsilon_{t+1}] = 0, \quad \text{Var}(\varepsilon_{t+1}) = I,$$

with  $\{\varepsilon_t\}$  serially uncorrelated over time, and define  $\Sigma_{g,r} := L_r L_r^\top$ .

### H-step Increment (Telescoping)

$$g_{t+H} - g_t = \sum_{h=1}^H (\mu_r + L_r \varepsilon_{t+h}) = H \mu_r + L_r \sum_{h=1}^H \varepsilon_{t+h}.$$

## Expectation and Variance

The constant  $H \mu_r$  has zero variance, hence

$$\text{Var}(g_{t+H} - g_t) = \text{Var}\left(L_r \sum_{h=1}^H \varepsilon_{t+h}\right) = L_r \text{Var}\left(\sum_{h=1}^H \varepsilon_{t+h}\right) L_r^\top.$$

Because the shocks are serially uncorrelated and  $\text{Var}(\varepsilon_{t+h}) = I$ ,

$$\text{Var}\left(\sum_{h=1}^H \varepsilon_{t+h}\right) = \sum_{h=1}^H \text{Var}(\varepsilon_{t+h}) + 2 \sum_{1 \leq h < \ell \leq H} \text{Cov}(\varepsilon_{t+h}, \varepsilon_{t+\ell}) = H I.$$

Therefore,

$$\text{Var}(g_{t+H} - g_t) = L_r (H I) L_r^\top = H L_r L_r^\top = H \Sigma_{g,r}.$$

## Implication for Long-Horizon Return

With  $Y_{i,H} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \varepsilon_{i,H}$ ,

$$\mathbb{E}[Y_{i,H}] = H \lambda_{i,r}^\top \mu_r, \quad \text{Var}(Y_{i,H}) \approx H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1),$$

i.e., the drift contributes to the mean while the variance scales linearly with  $H$  via  $\Sigma_{g,r}$ .

## Derivation of $\mathbb{E}[Y_{i,H}] = H \lambda_{i,r}^\top \mu_r$

We have

$$Y_{i,H} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \varepsilon_{i,H}, \quad \varepsilon_{i,H} = \Gamma_i^\top (z_{t+H} - z_t) + (u_{i,t+H} - u_{i,t}).$$

Within regime  $r$ , the global factors follow

$$g_{t+1} = g_t + \mu_r + L_r \varepsilon_{t+1}, \quad \mathbb{E}[\varepsilon_{t+1}] = 0.$$

Telescoping yields

$$g_{t+H} - g_t = H \mu_r + L_r \sum_{h=1}^H \varepsilon_{t+h} \Rightarrow \mathbb{E}[g_{t+H} - g_t] = H \mu_r.$$

Since  $z_t$  and  $u_{i,t}$  are stationary with constant means,  $\mathbb{E}[z_{t+H} - z_t] = 0$  and  $\mathbb{E}[u_{i,t+H} - u_{i,t}] = 0$ , hence  $\mathbb{E}[\varepsilon_{i,H}] = 0$ . Therefore,

$$\mathbb{E}[Y_{i,H}] = \lambda_{i,r}^\top \mathbb{E}[g_{t+H} - g_t] + \mathbb{E}[\varepsilon_{i,H}] = \lambda_{i,r}^\top (H \mu_r) = H \lambda_{i,r}^\top \mu_r.$$

With  $\varepsilon_{i,H}$  uncorrelated with  $g_t$ ,

$$\text{Var}(Y_{i,H}) = H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1).$$

## Where the Variance Expression Comes From

From the  $H$ -period return

$$Y_{i,H} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \varepsilon_{i,H}, \quad \text{write } \Delta_H g := g_{t+H} - g_t.$$

Use  $\text{Var}(A + B) = \text{Var}(A) + \text{Var}(B) + 2 \text{Cov}(A, B)$ :

$$\text{Var}(Y_{i,H}) = \text{Var}(\lambda_{i,r}^\top \Delta_H g) + \text{Var}(\varepsilon_{i,H}) + 2 \text{Cov}(\lambda_{i,r}^\top \Delta_H g, \varepsilon_{i,H}).$$

Because  $\lambda_{i,r}$  is constant,

$$\text{Var}(\lambda_{i,r}^\top \Delta_H g) = \lambda_{i,r}^\top \text{Var}(\Delta_H g) \lambda_{i,r}.$$

Using

$$\text{Var}(\Delta_H g) = H \Sigma_{g,r}.$$

Hence

$$\text{Var}(\lambda_{i,r}^\top \Delta_H g) = H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r}.$$

Next,  $\varepsilon_{i,H} = \Gamma_i^\top (z_{t+H} - z_t) + (u_{i,t+H} - u_{i,t})$  is a difference of stationary processes, so

$$\text{Var}(\varepsilon_{i,H}) = O(1) \quad (\text{bounded in } H).$$

Under the stated orthogonality assumption,  $\text{Cov}(\Delta_H g, \varepsilon_{i,H}) = 0$ , so the cross term vanishes. Therefore,

$$\boxed{\text{Var}(Y_{i,H}) = H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1).}$$

Hence the share of total variance explained by global factors at horizon  $H$  is

$$\frac{\text{factor variance}}{\text{total variance}} = \frac{H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r}}{H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1)} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r}}{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1/H)}.$$

**Note:** Factor variance is defined as the variance contribution to asset  $i$ 's dynamics that comes from exposure to common factor increments over horizon  $H$ . By *common factor increments* we mean the net change in the global factor process  $g_t$  over a horizon  $H$ ,

$$\Delta_H g = g_{t+H} - g_t.$$

This captures how much the common factors move over that time window. The variance of these increments,

$$\text{Var}(\Delta_H g) = H \Sigma_{g,r},$$

scales linearly with  $H$ . When projected onto the loadings  $\lambda_{i,r}$ , this gives the part of asset  $i$ 's variance driven by the change in global shocks. For fixed  $H$ , an increase in the factor covariance  $\Sigma_{g,r}$  (typical in crises) *raises* this ratio; for large  $H$ ,  $O(1/H) \rightarrow 0$  and the factor share tends to 1. This is the **variance channel**: higher global factor volatility makes common shocks dominate long-horizon variance.

## Cross-Currency Correlation (“Loading-Alignment Channel”)

For currencies  $i$  and  $j$ ,

$$\text{Cov}(Y_{i,H}, Y_{j,H}) = H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1),$$

### Derivation of the Covariance Term

From the  $H$ -period return

$$Y_{i,H} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \varepsilon_{i,H}, \quad Y_{j,H} = \lambda_{j,r}^\top (g_{t+H} - g_t) + \varepsilon_{j,H},$$

the covariance is

$$\text{Cov}(Y_{i,H}, Y_{j,H}) = \text{Cov}\left(\lambda_{i,r}^\top (g_{t+H} - g_t) + \varepsilon_{i,H}, \lambda_{j,r}^\top (g_{t+H} - g_t) + \varepsilon_{j,H}\right).$$

### Reminder (Bilinearity of Covariance)

For random variables  $A, B, C, D$ ,

$$\text{Cov}(A + B, C + D) = \text{Cov}(A, C) + \text{Cov}(A, D) + \text{Cov}(B, C) + \text{Cov}(B, D).$$

Applying this property yields

$$\begin{aligned} \text{Cov}(Y_{i,H}, Y_{j,H}) &= \text{Cov}(\lambda_{i,r}^\top (g_{t+H} - g_t), \lambda_{j,r}^\top (g_{t+H} - g_t)) \\ &\quad + \text{Cov}(\varepsilon_{i,H}, \varepsilon_{j,H}) + \text{cross terms.} \end{aligned}$$

By the orthogonality in our setup,  $\text{Cov}(\Delta_H g, \varepsilon_{i,H}) = \text{Cov}(\Delta_H g, \varepsilon_{j,H}) = 0$ , so both cross terms vanish:

$$\text{Cov}(\lambda_{i,r}^\top \Delta_H g, \varepsilon_{j,H}) = \text{Cov}(\varepsilon_{i,H}, \lambda_{j,r}^\top \Delta_H g) = 0.$$

By assumption,  $\varepsilon_{i,H}$  and  $\varepsilon_{j,H}$  are differences of stationary processes with bounded variance  $O(1)$ , and they are uncorrelated with the factor increment  $g_{t+H} - g_t$ . Thus the leading term is

$$\text{Cov}(\lambda_{i,r}^\top (g_{t+H} - g_t), \lambda_{j,r}^\top (g_{t+H} - g_t)).$$

Since  $\lambda_{i,r}$  and  $\lambda_{j,r}$  are constant vectors,

$$\text{Cov}(\lambda_{i,r}^\top (g_{t+H} - g_t), \lambda_{j,r}^\top (g_{t+H} - g_t)) = \lambda_{i,r}^\top \text{Var}(g_{t+H} - g_t) \lambda_{j,r}.$$

Using the variance scaling result  $\text{Var}(g_{t+H} - g_t) = H \Sigma_{g,r}$ , we obtain

$$\text{Cov}(Y_{i,H}, Y_{j,H}) = H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1).$$

$$\text{Var}(Y_{i,H}) = H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1), \quad \text{Var}(Y_{j,H}) = H \lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1).$$

Therefore the correlation at horizon  $H$  is

$$\rho_{ij}^{(H)} = \frac{H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1)}{\sqrt{(H \lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1))(H \lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1))}} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1/H)}{\sqrt{(\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r} + O(1/H))(\lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r} + O(1/H))}}.$$

Letting  $H \rightarrow \infty$  yields the limit

$$\lim_{H \rightarrow \infty} \rho_{ij}^{(H)} = \frac{\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{j,r}}{\sqrt{(\lambda_{i,r}^\top \Sigma_{g,r} \lambda_{i,r})(\lambda_{j,r}^\top \Sigma_{g,r} \lambda_{j,r})}} = \cos(\angle(\Sigma_{g,r}^{1/2} \lambda_{i,r}, \Sigma_{g,r}^{1/2} \lambda_{j,r})).$$

This is the cosine of the angle between the vectors  $\Sigma_{g,r}^{1/2} \lambda_{i,r}$  and  $\Sigma_{g,r}^{1/2} \lambda_{j,r}$ . If loadings *align* across currencies ( $\lambda_{i,r} \approx \lambda_{j,r}$ ), then the cosine is close to 1, implying near-perfect long-run correlation. This is the **loading-alignment channel**.

## Interpretation and crisis amplification

During crises, Cross-currency responses to global shocks often become more similar (compression/alignment of  $\lambda_{i,r}$  across  $i$ ), pushing the cosine above closer to 1 (loading-alignment channel).

Together these mechanisms imply stronger co-movement of long-horizon currency returns in crises: common factors explain a larger fraction of variance, and correlations across currencies rise toward the cosine bound, which approaches 1 when loadings align.

## Proposition 3 (Rank Change in Crisis and Cointegration)

Consider the factor structure of currency returns before and during crisis regimes:

$$X_t = \mu_r + \Lambda_r g_t + \Gamma_r z_t + u_t,$$

where:

- $X_t$  is the  $N \times 1$  vector of currency returns,
- $\Lambda_r$  is the  $N \times K_r$  loading matrix on persistent global factors  $g_t$ ,

- $g_t$  is a  $K_r \times 1$  vector of persistent global factors (stochastic trends),
- $\Gamma_r z_t + u_t$  is a stationary component,
- The regime  $r \in \{\text{pre-crisis, crisis}\}$  indexes different periods.

#### Relation between rank of $\Lambda_r$ and cointegration

By standard results in cointegration theory (see Johansen 1991; Stock and Watson 1988), the number of common stochastic trends driving  $X_t$  equals the rank of  $\Lambda_r$ , denoted  $\text{rank}(\Lambda_r) = K_r$ . The dimension of the cointegrating space, i.e., the number of stationary linear combinations of  $X_t$ , equals

$$\dim(\text{cointegrating space}) = N - K_r.$$

#### Effect of new persistent global factors

Suppose that during crisis, new persistent global factors emerge that are independent of the pre-crisis factors. Formally, the crisis regime factor vector can be written as

$$g_t^{\text{crisis}} = \begin{bmatrix} g_t^{\text{pre}} \\ g_t^{\text{new}} \end{bmatrix},$$

where  $\dim(g_t^{\text{new}}) = K_{\text{new}} > 0$ , and  $g_t^{\text{new}}$  is independent of  $g_t^{\text{pre}}$ . The corresponding loading matrix expands as

$$\Lambda_{\text{crisis}} = [\Lambda_{\text{pre}} \quad \Lambda_{\text{new}}],$$

with  $\Lambda_{\text{new}}$  loading persistently on the new factors. Because  $g_t^{\text{new}}$  is independent and  $\Lambda_{\text{new}}$  has full column rank  $K_{\text{new}}$ , the rank of  $\Lambda_{\text{crisis}}$  satisfies

$$\text{rank}(\Lambda_{\text{crisis}}) \geq \text{rank}(\Lambda_{\text{pre}}) + K_{\text{new}}.$$

Thus,

$$K_{\text{crisis}} = \text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}) = K_{\text{pre}}.$$

### Effect on cointegration relations

Since the number of cointegrating relations equals  $N - K_r$ , an increase in the rank  $K_r$  reduces the cointegration dimension:

$$N - K_{\text{crisis}} < N - K_{\text{pre}}.$$

In the extreme case where

$$K_{\text{crisis}} = N,$$

the number of cointegrating relations becomes zero, meaning no stationary linear combination exists — the system has no cointegration.

### Effect of autonomy shifts on rank

Alternatively, even without new factors, if autonomy shifts such that the loading matrix  $\Lambda$  changes rank (e.g., if some currency responses become linearly independent from others), then

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}),$$

and the same conclusion on cointegration applies.

**Conclusion:** The emergence of new persistent global factors or autonomy shifts that increase the rank of  $\Lambda$  during crisis reduce the number of cointegrating relations, and if the rank reaches  $N$ , cointegration disappears entirely.  $\square$

## Explanation of Proposition 3 (Rank Change in Crisis and Cointegration)

### Setup: Factor Structure

We start with the factor representation of currency returns:

$$X_t = \mu_r + \Lambda_r g_t + \Gamma_r z_t + u_t,$$

where:



- $X_t$  is the  $N \times 1$  vector of currency returns,
- $\Lambda_r$  is the  $N \times K_r$  loading matrix on persistent global factors,
- $g_t$  is the  $K_r \times 1$  vector of persistent global factors (stochastic trends),
- $\Gamma_r z_t + u_t$  is the stationary component,
- $r \in \{\text{pre-crisis, crisis}\}$  indicates the regime.

## Rank and Cointegration Relationship

By standard results from cointegration theory, the rank of the loading matrix  $\Lambda_r$  equals the number of independent stochastic trends:

$$K_r = \text{rank}(\Lambda_r).$$

The dimension of the cointegrating space, i.e., the number of stationary linear combinations of  $X_t$ , is therefore:

$$\dim(\text{cointegrating space}) = N - K_r.$$

## Crisis Introduces New Global Factors

During a crisis, new persistent global factors emerge that are independent of the pre-crisis factors. Formally, the factor vector expands to:

$$g_t^{\text{crisis}} = \begin{bmatrix} g_t^{\text{pre}} \\ g_t^{\text{new}} \end{bmatrix},$$

where  $\dim(g_t^{\text{new}}) = K_{\text{new}} > 0$  and  $g_t^{\text{new}}$  is independent of  $g_t^{\text{pre}}$ . The loading matrix correspondingly expands to:

$$\Lambda_{\text{crisis}} = [\Lambda_{\text{pre}} \quad \Lambda_{\text{new}}],$$

with  $\Lambda_{\text{new}}$  loading on the new factors. Because  $g_t^{\text{new}}$  is independent and  $\Lambda_{\text{new}}$  has full column rank  $K_{\text{new}}$ , the rank of the loading matrix increases:

$$\text{rank}(\Lambda_{\text{crisis}}) \geq \text{rank}(\Lambda_{\text{pre}}) + K_{\text{new}}.$$

Thus, the number of stochastic trends in the crisis regime is strictly higher:

$$K_{\text{crisis}} = \text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}) = K_{\text{pre}}.$$

## Effect on Cointegration

Since the number of cointegration relations equals  $N - K_r$ , an increase in  $K_r$  reduces the cointegration dimension:

$$N - K_{\text{crisis}} < N - K_{\text{pre}}.$$

In the extreme case where:

$$K_{\text{crisis}} = N,$$

we have:

$$N - K_{\text{crisis}} = 0,$$

meaning that there are no stationary linear combinations of  $X_t$ , and the system has no cointegration.

## Autonomy Shifts (Without New Factors)

Even if no new factors appear, the rank can still increase if existing currency returns become less dependent on each other (i.e., the loading matrix  $\Lambda$  gains linear independence among its columns). In this case:

$$\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}}),$$

and the same conclusion applies: the dimension of the cointegrating space decreases.

## Conclusion

The emergence of new persistent global factors or structural shifts that increase the rank of  $\Lambda$  during crisis periods reduces the number of cointegrating relations:

$$N - K_{\text{crisis}} < N - K_{\text{pre}}.$$

If the rank reaches  $N$ , then:

- All currencies are driven by independent stochastic trends, and
- No cointegration exists in the system.

Thus, crises lead to weaker cointegration among currency returns as the system becomes more dominated by independent global factors.